

Smart Sweep SWMS

Requirements gap analysis against the KCC stakeholder document

Prepared 10 August 2026 · Source: Documents.docx (KCC, primary collectors, data management) · Reviewed against the live application

How to read this

Every requirement in the source document has been checked against the application as it actually stands — the database models, the API routes and the screens — rather than against a feature list. Each item is marked:

Built In the product and usable today	Partial The foundation exists; something specific is missing	Missing Not addressed at all
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The headline: the household and property registry, billing and money trail, complaints, routes, vehicles and the new survey module are substantially built. The gaps cluster in three places — anything that has to leave the system and reach a person (notifications, campaigns, published tariffs), anything about the workforce beyond who they are (attendance, training, PPE, health, salary), and anything about service geography finer than a ward (road-wise operator assignment, dumping points, bins).

Summary

22 requirement areas assessed: 5 built, 11 partial, 6 missing.

Requirement	Status	Principal gap
A1. Monthly reporting and accountability	Partial	There is no monthly return as a document an operator submits and KCC marks off.
A2. Road-wise service coverage and authorisation	Missing	Service areas are defined at ward level only (Agency.service_wards).
A3. Household coverage and registry	Built	Only the campaign side is absent: uncovered households can be listed but not worked as a campaign list (see B2/B6).
B1. Service continuity and backup arrangements	Missing	Attendance is today's value only — there is no dated register, so "how often was this collector absent last month" cannot be answered.
B2. Citizen campaign and enforcement support	Missing	No campaign module of any kind: miking, handbill distribution, courtyard meetings and ward campaigns have nowhere to be recorded or tracked.
B3. Daily work records and complaint handling	Built	A complaint is located by ward and household rather than by a map pin.
B4. Collection delay and missed-collection alerts	Partial	Nothing raises an alert.
B5. Infrastructure, equipment and dumping points	Partial	Everything that is not a vehicle is absent.
B6. Community participation and engagement	Missing	No engagement tracker: no contact history, no participation status over time, no campaign coverage per household and no follow-up actions.
B7. Financial sustainability and fee collection	Partial	No formal receipt is issued to the household — the payment reference field is a bKash or bank transaction id, not a numbered receipt the citizen keeps.
B8. Worker capacity, training and wellbeing	Missing	None of the wellbeing or capacity data the document asks for exists: no training record, no attendance register, no salary register, no health incident log, no PPE issue and status tracking, and no document storage for appointment letters.

Requirement	Status	Principal gap
B9. Monitoring, escalation and taskforce	Partial	The SLA is recorded but not enforced — nothing escalates a breach or alerts anyone.
C1. Baseline and household information	Built	—
C2. Waste worker and route records	Partial	Appointment letters, salary register, monthly worker meeting records and training records have no home in the system.
C3. Finance and payment records	Partial	Cash book, ledger and voucher records are not modelled, and no fee collection sheet or numbered receipt is produced as a document.
C4. Vehicle, equipment and stock records	Partial	Stock register, equipment deficit records, and disposal or dispatch records are absent, as is any dumping point record.
C5. Service, complaint and reporting records	Partial	Meeting resolutions, training and workshop files, and a knowledge repository have nowhere to live — the document lists all three as records KCC keeps today.
Cross-cutting: Bengali language support	Built	The whole interface ships in Bangla and English, and a test enforces that every key exists in both so a new screen cannot ship half-translated.
Cross-cutting: Role-based access	Built	Four roles, with access scoped on two independent axes — ward and agency — so two contractors working the same ward never see each other's records.
Cross-cutting: Mobile-friendly for low-skill users	Partial	The layout is responsive and the daily round screen is built for a phone, with QR scanning for households.
Cross-cutting: Working offline in the field	Partial	The server side is ready: visits and surveys both accept a queued batch and answer per row, so one bad record cannot cost a worker their morning.
Cross-cutting: Notifications to citizens and staff	Missing	Nothing leaves the system.

Findings in detail

A. Khulna City Corporation

A1. Monthly reporting and accountability **PARTIAL**

What exists. Agency master holds each operator with type (private / NGO / CBO / cooperative), contract dates, licence and TIN/BIN. Royalty is tracked end to end: collector deposit → agency → KCC remittance, with a reconciliation report that ties collected, deposited and remitted together per month. Reports for service delivery, bill collection, complaints and KPIs all filter by agency.

What is missing. There is no monthly return as a document an operator submits and KCC marks off. Nothing answers “which operators have reported for July, and which have not”. The figures exist but are pulled by KCC, not pushed by the operator, so an operator that stops reporting is invisible rather than flagged. No exception alerts. “Individual collector” is not one of the four agency types.

A2. Road-wise service coverage and authorisation **MISSING**

What exists. Roads exist as real records under each ward. Routes carry an ordered stop list and are stamped with the agency that walks them, and every holding can hold a verified GPS pin. A Leaflet map is already in the product for live vehicle tracking.

What is missing. Service areas are defined at ward level only (Agency.service_wards). There is no road-to-operator assignment, so the question the document asks — “which NGO or operator works which road” — cannot be answered directly. No written authorisation is recorded against a service area. Nothing detects two operators claiming the same road, or an operator working outside their area. The map shows vehicles, not service boundaries or coverage status.

A3. Household coverage and registry **BUILT**

What exists. This is the most complete area. Holding Master records the building (holding number, owner name and mobile, ward, road, GPS pin, type, unit count) and households hang off it, so one building with twelve flats is one address and twelve billable customers. Service status is derived from the households rather than stored, so it cannot go stale. Households outside the service are held separately as potential customers with a recorded reason. Collection history is the visit log, per household per day.

What is missing. Only the campaign side is absent: uncovered households can be listed but not worked as a campaign list (see B2/B6).

B. Primary Collector

B1. Service continuity and backup arrangements **MISSING**

What exists. Collectors carry a current status and attendance value, and employment history records which agency employed them and when.

What is missing. Attendance is today’s value only — there is no dated register, so “how often was this collector absent last month” cannot be answered. No backup worker arrangement: nothing names a substitute or reassigns a route when someone is sick, which is the specific failure the document describes. Supervisor and collector contact numbers are not published to holding owners.

B2. Citizen campaign and enforcement support **MISSING**

What exists. Service tiers carry the standard monthly charge, so tariff figures exist. Households outside the service are already listed with a reason and current disposal practice.

What is missing. No campaign module of any kind: miking, handbill distribution, courtyard meetings and ward campaigns have nowhere to be recorded or tracked. No violation or enforcement register for illegal disposal, and no follow-up workflow. Tariff information is internal only — nothing publishes it to citizens.

B3. Daily work records and complaint handling BUILT

What exists. Complaints carry a ticket id, type, ward, household, channel, priority, assigned person, status and SLA, with a full activity trail and photo attachments. Daily work is the visit log: every household visit records collector, time, GPS accuracy, whether it was scanned or keyed, and a skip reason when waste was not taken. Both survive a poor connection through bulk upload endpoints that report per-row outcomes.

What is missing. A complaint is located by ward and household rather than by a map pin. There is no daily work register as a signed-off document per collector per day — the data is there, but as a derived view rather than a submitted return.

B4. Collection delay and missed-collection alerts PARTIAL

What exists. The daily round screen shows each collector their stops in walking order with counts for done, skipped and pending. Coverage and on-time percentages are recalculated from the visit log rather than typed in.

What is missing. Nothing raises an alert. A round that is never completed produces a low number on a report somebody has to open — there is no missed-collection alert, no route-completion signal and no supervisor verification step. Waste found dumped on a road inside the coverage area has nowhere to be recorded.

B5. Infrastructure, equipment and dumping points PARTIAL

What exists. Vans are fully covered: plate, type, capacity, fuel, ownership, GPS unit, odometer, driver, and expiry dates for fitness, tax, insurance and permit, plus maintenance and fuel logs and live positions.

What is missing. Everything that is not a vehicle is absent. No dumping points or secondary transfer stations, no bin inventory or bin location map, no equipment or stock register, and no record of equipment deficits — all of which the document names explicitly. Van condition is a single status value, not an inspection record.

B6. Community participation and engagement MISSING

What exists. Household and holding records carry owner contact details, and the new survey module captures willingness to pay, current disposal practice and reasons for not using a van.

What is missing. No engagement tracker: no contact history, no participation status over time, no campaign coverage per household and no follow-up actions. Local committees have nowhere to be recorded. The survey answers the question once; nothing tracks what was done about the answer.

B7. Financial sustainability and fee collection PARTIAL

What exists. Strong coverage. Monthly billing runs generate bills per household; payments record amount, method, collector, agency and transaction reference; arrears and dues are reported per household and per ward; and money is traceable from the doorstep to KCC through deposit and remittance with a reconciliation report.

What is missing. No formal receipt is issued to the household — the payment reference field is a bKash or bank transaction id, not a numbered receipt the citizen keeps. No cash book, ledger or voucher views, which the document lists as current practice. Households repeatedly unavailable at collection time are not flagged.

B8. Worker capacity, training and wellbeing MISSING

What exists. Collector records hold name, staff id, phone, licence number and expiry, joining date and current agency, with employment history across agencies.

What is missing. None of the wellbeing or capacity data the document asks for exists: no training record, no attendance register, no salary register, no health incident log, no PPE issue and status tracking, and no document storage for appointment letters. This is the largest single gap by volume of missing records.

B9. Monitoring, escalation and taskforce PARTIAL

What exists. Complaints carry priority and an SLA in hours, and the dashboard gives supervisors open counts by status and ward. Access is role-based and scoped by ward and agency, so each operator sees only their own work.

What is missing. The SLA is recorded but not enforced — nothing escalates a breach or alerts anyone. No taskforce issue tracker separate from citizen complaints, so an internal operational problem has to be filed as if a citizen reported it. No action-status monitoring or escalation workflow.

C. Data Management

C1. Baseline and household information **BUILT**

What exists. Location, ward, holding number, owner details, mobile number and the number of households under each holding are all held, and the holding is the single writer of the address so the figures cannot disagree between screens.

C2. Waste worker and route records **PARTIAL**

What exists. Route assignment is complete: routes, ordered stops, assignments to collectors, and reassignment. Collector biodata is partly held.

What is missing. Appointment letters, salary register, monthly worker meeting records and training records have no home in the system. Attendance is current-state only.

C3. Finance and payment records **PARTIAL**

What exists. Fee collection, service charge communication and royalty payment records are covered by bills, payments, deposits and remittances with a reconciliation view.

What is missing. Cash book, ledger and voucher records are not modelled, and no fee collection sheet or numbered receipt is produced as a document.

C4. Vehicle, equipment and stock records **PARTIAL**

What exists. Van register and van maintenance register are complete and in daily use.

What is missing. Stock register, equipment deficit records, and disposal or dispatch records are absent, as is any dumping point record.

C5. Service, complaint and reporting records **PARTIAL**

What exists. Service recipient register, complaint register and the reporting dashboard are built, and the survey data collection sheet is now a full versioned survey module with a 72-question door-to-door form, offline capture and promotion of a reviewed survey into the holding register.

What is missing. Meeting resolutions, training and workshop files, and a knowledge repository have nowhere to live — the document lists all three as records KCC keeps today.

Cross-cutting requirements

The document closes by asking for a system that is mobile-friendly, Bengali-supported, role-based and introduced gradually to workers with limited digital skills. Those are assessed separately here because they cut across every module.

Bengali language support **BUILT**

The whole interface ships in Bangla and English, and a test enforces that every key exists in both so a new screen cannot ship half-translated. The survey module carries Bangla for every question, hint and option, served from the database rather than the front end, so a revised form needs no release.

Role-based access **BUILT**

Four roles, with access scoped on two independent axes — ward and agency — so two contractors working the same ward never see each other's records. Write guards apply on the way in as well as the way out.

Mobile-friendly for low-skill users **PARTIAL**

The layout is responsive and the daily round screen is built for a phone, with QR scanning for households. But this is a browser application, not an installable app: there is no home-screen install, no offline shell, and a worker who loses signal loses the page.

Working offline in the field **PARTIAL**

The server side is ready: visits and surveys both accept a queued batch and answer per row, so one bad record cannot cost a worker their morning. The design the document needs is in place.

Notifications to citizens and staff **MISSING**

Nothing leaves the system. There is no SMS or email delivery at all — the login code is generated but never sent, and SMS exists only as a label on how a complaint arrived. Tariff notices, collection reminders, dues reminders and complaint updates all require this.

Recommended order of work

Ordered by what the document treats as urgent, weighted towards work that unblocks several gaps at once and that builds on what is already there. Effort is relative to the modules already delivered, not an estimate in days.

1. Road-wise operator assignment and authorisation (A2)

KCC's clearest unmet ask, and everything about overlap and unauthorised entry depends on it. Assign roads to operators, record the authorisation, and flag roads claimed twice or served by nobody.

Effort: Medium — the road and agency records already exist; this is a join, a screen and an overlap report.

2. Monthly operator return with submission tracking (A1)

Turns reporting from something KCC pulls into something operators submit and KCC signs off, which is what accountability means here. The figures are already calculated; what is missing is the envelope and the submitted / not submitted state.

Effort: Medium — mostly assembling existing reports behind a submit-and-review flow.

3. SMS notifications (Cross-cutting)

Unblocks tariff communication, dues reminders, complaint updates and collection alerts in one piece of work. Several other gaps are only half-solvable without it. A provider is already in use elsewhere in your estate.

Effort: Small — one service, one queue, and the call sites.

4. Worker register: attendance, training, PPE, health, salary (B8)

The largest block of missing records, and the document is specific about each one. Also the precondition for the backup-worker arrangement in B1.

Effort: Large — several related records, but each one simple.

5. Missed-collection alerts and supervisor verification (B4)

The visit data needed already exists; what is missing is the rule that turns an incomplete round into something that reaches a supervisor the same day.

Effort: Small — a scheduled check over the visit log, plus a verification action.

6. Campaign and enforcement module (B2 / B6)

Covers miking, handbills, courtyard meetings, ward campaigns, the non-participating household list and violation follow-up. Directly serves the behaviour problem the collectors described.

Effort: Medium — a campaign record, a participation tracker and a violation register.

7. Dumping points, bins and equipment stock (B5 / C4)

Completes the asset side, which currently stops at vehicles.

Effort: Medium — three straightforward registers plus a bin location map.

8. Numbered receipts, cash book and voucher trail (B7 / C3)

Matches the paper records operators keep today, which matters for adoption and for audit.

Effort: Medium — a receipt series on payments, then ledger views over existing data.

9. Taskforce tracker and SLA escalation (B9)

Separates internal operational issues from citizen complaints, and makes the SLA that is already recorded actually do something.

Effort: Small to medium — escalation rules over complaints, plus an internal issue type.

10. Installable mobile app with offline capture (Cross-cutting)

The server already accepts queued batches; the missing half is a client that keeps working with no signal. The document is explicit that workers have limited digital skills and need this to be simple.

Effort: Large — offline storage, sync state and an install target.

A note on what is already strong

The document's stated first phase is "household registry, road-wise operator mapping, daily work reporting, complaint management, fee and royalty tracking, workforce and van records, and a KCC monitoring dashboard". Six of those seven are built or close to it. Road-wise operator mapping is the one that is not started, and workforce records are the one that is thinnest — which is why they sit first and fourth in the list above.

The survey module delivered most recently also covers the document's "survey data collection sheet" requirement more thoroughly than asked: the questionnaire is stored as data rather than code, so a revised form is a configuration change, and a reviewed survey can be promoted straight into the holding register.

Prepared from the source requirements document and a direct review of the application code.